Revisiting the Intra-Team Communication Method to Elicit Level-k Reasoning in Beauty Contests and 11–20 Games*

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Extended Abstract

In behavioral game theory, the "Level-k" model (Nagel, 1995; Stahl and Wilson, 1995) has become a workhorse solution concept for organizing widely observed non-equilibrium behavior in strategic environments. Rather than assuming mutually consistent beliefs, the Level-k model posits a hierarchy of heterogeneous sophistication: level-0 players are non-strategic, while each level-k player ($k \geq 1$) believes others are level $k-1$ and best respond accordingly. As a result, the model's predictions hinge on how level-0 players' behavior is specified. However, both their actual behavior and strategic players' beliefs about them remain important open questions, motivating approaches that elicit reasoning beyond observed choices.

To this end, Burchardi and Penczynski (2014) propose the *intra-team communication* method, in which players make strategic decisions as a team. Before deciding, players are incentivized to send a message to "convince" their teammate, allowing experimenters to infer their beliefs and reasoning. First applied to beauty contest games, a cornerstone of the level-k literature, this method represents a key methodological innovation in the experimental study of strategic behavior, providing one of the few approaches to directly elicit underlying reasoning and being widely used across multiple domains.[1] Despite its impact, the original application in the beauty contest game has never been replicated.

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[1] The intra-team communication method has been applied in various contexts, including hide-and-seek

In this paper, we provide the first replication of the original beauty contest experiment with the intra-team communication method of Burchardi and Penczynski (2014). We also apply this method to the 11–20 game, another canonical setting for studying level-k reasoning proposed by Arad and Rubinstein (2012). Focusing on the "re-ordered" version introduced by Goeree et al. (2018), which exhibits heterogeneous noisy behavior not well explained by the standard level-k model, we not only provide the first replication of this re-ordered 11–20 game, but also use the method to illuminate the reasoning underlying such behavior and its relationship to level-k thinking. Overall, our study offers an integrated perspective on empirical patterns in two canonical games with boundedly rational behavior.

JEL Classification Number: C81, C91, D83

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(Penczynski, 2016b), persuasion (Penczynski, 2016a), social learning (Penczynski, 2017), coordination games (van Elten and Penczynski, 2020), contests (Bruner et al., 2022) and the Colonel Blotto resource allocation game (Arad et al., 2024; Arad and Penczynski, 2024).

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Appendix A Experimental Instructions[2]

General Instructions

You are about to participate in an experiment in team decision making. Your earnings will depend partly on your decisions, partly on the decision of others, and partly on chance.

Please turn off all your mobile devices. The entire session will take place through computer terminals, and all interactions between participants will be conducted through the computers. Do not open any applications on the computer unless you are instructed to do so. Do not talk or in any way try to communicate with other participants during the session, except according to our instructions.

The experiment consists of two parts (**Part I** and **Part II**), which are independent of each other and feature different tasks, with three rounds in each part. You will receive the instructions for Part II after Part I has concluded. If you have any questions during the instructions, please raise your hand. The experimenter will approach you and answer your question confidentially. However, if the question appears relevant to others or reflects a common misunderstanding, it will be answered so that everyone can hear, for clarity and transparency.

Since this is a team experiment, you will at various times be matched randomly with another participant in this room, to form a team that plays as one entity. Your team's earnings will be shared equally between you and your team partner. The way you interact as a team to take decisions will be the same throughout the two parts.

Now, let us explain how your ***Team's Action*** is determined. In each round, both your team partner and you will enter a ***Final Decision*** individually and the computer will choose randomly which one of your two final decisions counts as your team's action. The probability that your team partner's final decision is chosen is equal to the probability that your final decision will be chosen (i.e. your chances are 50:50). However, you have the possibility to influence your partner's final decision in the following way: Before you enter your final decision, you can propose to your partner a ***Suggested Decision*** and send her/him one and only one text ***Message***. *Note that this message is your only chance to convince your partner of the reasoning behind your suggested decision. Therefore, use the message to explain your suggested decision to your team partner.* After you finish entering your suggested decision and your message, these will be shown to your team partner. She/he will then make her/his final decision. Similarly, you will receive your partner's suggested decision and message. You will then make your final decision. As outlined above, once you both enter your final decision, the computer chooses randomly one of your final decisions as your team's action.

In this experiment, you will earn "points" in each round. Your earnings will be determined by the sum of all the points you earn across both parts. Each point has a value of 25 cents.

[2] This appendix provides the experimental instructions for the treatment order in which subjects first play three rounds of the beauty contest game, followed by three rounds of the 11–20 game. The instructions for the other treatment order are identical, except that Parts One and Two are swapped.

That is, every 4 points generates $1 in earnings for you. In addition to your earnings from decisions, you will receive a show-up fee of $10. At the end of the session, we will record your earnings and you will be paid via a UVA-issued electronic check through PaymentWorks. Our department staff will email you with payment details.

In order for you to get familiar with the messaging system, you will now try it out in a **Test Period**.

Test Period

A participant in this room is now randomly chosen to be your team partner. The **Test Period** has two rounds, with one question to answer in each round. Since this is only a test, your earnings will **not** depend on any decision taken now. In both test rounds you will need to answer a question about the year of a historical event. The team that is closest to the correct year wins.

As described, you will be able to send one ***Suggested Decision*** with your proposed year and an explaining ***Message***. After having read your partner's suggested decision and message, you will enter your ***Final Decision***. As described earlier, either your or your partner's final decision will be chosen randomly to be your ***Team's Action***.

You can send ***Messages*** of any length by typing in the text box on the screen. You can delete any part of your message using the "Backspace" key on your keyboard. Once you have finished typing your message, please click the "Send Message" button to send it to your team partner.

When you are ready, please click the "Ready" button to start the Test Period.

Start Part I

You are about to start **Part I** of the experiment. You are now randomly matched with a new team partner. For each of the next three rounds you will be matched with a new team partner, i.e. in each of the following rounds you will play with a different person.

In each round, once all final decisions have been taken, the computer will record which of the final decisions (yours or your team partner's) is chosen randomly as your team's action, whether your team won the round, and your personal earnings. **In each round, the winning team earns 100 points (50 points per team member).** You will be informed about all the records in this and the other part at the end of the experiment, but not between rounds.

Then the next round of the game follows. It will feature an identical task, but you will be matched with a new team partner.

Your task is the following:

Your team and all other teams will take their Team's Action by choosing a number between 0 and 100. 0 and 100 are also possible. Only whole numbers will be accepted. From all teams' actions, the computer will calculate the *Average*. A fraction p of this average will be your *target* number. The winning team will be the one that is *closest to p times the average*. For example, if p is two thirds (2/3), then the winning team will be the one that is closest to two thirds of the average. If two or more teams are equally close, the prize will be randomly given to one of these teams.

Important: The value of p may change from round to round. At the beginning of each round, the screen will display the value of p for that round. All participants will see the same value of p as you do.

As described earlier, you will send your team partner a ***Suggested Decision*** and a ***Message***. Remember to explain in the message your reasoning behind your suggested decision. After this information is exchanged, both of you enter your ***Final Decision***, from which the computer randomly chooses the ***Team's Action***.

You will be paid the sum of all points you earn in the three rounds (plus the points you earn in the other part).

When you click the "Ready" button, you will start the first round of **Part I** of the experiment.

Start Part II

You are about to start **Part II** of the experiment. You are now randomly matched with a new team partner. For each of the next three rounds you will be matched with a new team partner, i.e. in each of the following rounds you will play with a different person.

In each round, once all final decisions have been taken, the computer will record which of the final decisions (yours or your team partner's) is chosen randomly as your team's action, how many points your team received, and your personal earnings. You will be informed about all the records in this and the other part at the end of the experiment, but not between rounds.

Your task is the following:

Your team and another team in the experiment are randomly matched to play the following game.

13. 16. 20. 18. 19. 12. 15. 11. 17. 14

On your screen you will see 10 boxes in line, containing different amounts (like the ones above). Each team will take its *Team's Action* by individually selecting one of the 10 boxes. Each team will receive the amount in the box corresponding to its Team's Action. **A team**

will receive an additional amount of 20 points if its selected amount is exactly "one to the left" of the box that the other team chooses.

Example: Suppose the boxes are arranged as shown above, and your Team's Action is 12:

- If the other team chooses 17, your team earns 12 points and the other team earns 17.
- If the other team chooses 15, your team earns 12 + 20 = 32 points and the other team earns 15 points.
- If the other team also chooses 12, both teams earn 12 points.
- If the other team chooses 19, your team earns 12 points and the other team earns 19 + 20 = 39 points.

Important: You will play three rounds, and the order of the boxes may differ from the example above and may change from round to round. At the beginning of each round, the screen will display the order of the boxes for that round. All participants will see the boxes in the same order as you do.

As described earlier, you will send your team partner a ***Suggested Decision*** and a ***Message***. Remember to explain in the message the reasoning behind your suggested decision. After this information is exchanged, both of you enter your ***Final Decision***, from which the computer randomly chooses the ***Team's Action***.

You will be paid the sum of all points you earn in the three rounds (plus the points you earn in the other part).

When you click the "Ready" button, you will start the first round of **Part II** of the experiment.

Appendix B Screenshots

In this appendix, we provide screenshots of the experimental program for the treatment in which subjects play the beauty contest games first, followed by the 11–20 games.[3] Figures A1 to A3 present sample screenshots from the first round of the beauty contest game, where $p = 2/3$. In rounds 2 and 3, the multipliers are $1/3$ and $1/2$, respectively. Figures A4 to A6 present sample screenshots from the first round of the re-ordered 11–20 game. The configuration of the first round corresponds to the extreme version in Goeree et al. (2018), while rounds 2 and 3 correspond to the moderate and baseline versions, respectively. Finally, Figure A7 presents a sample screenshot of the feedback screen shown at the end of the experiment.

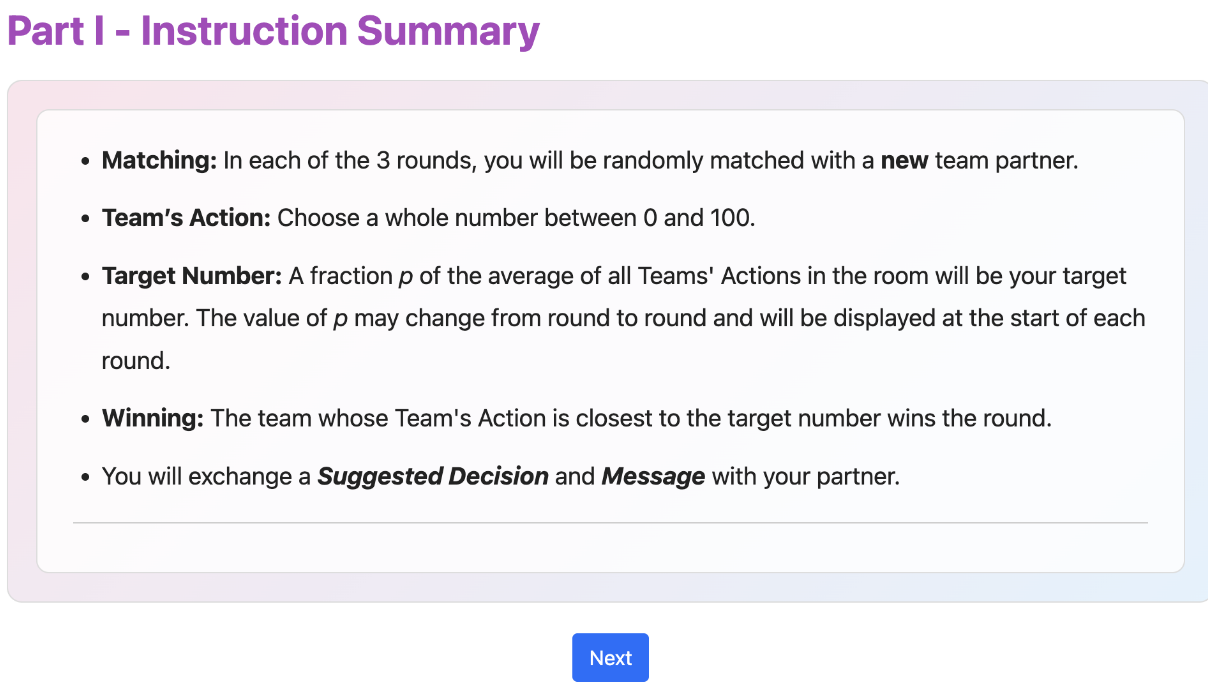

Figure A1: Instruction Summary of the Beauty Contest Game

[3] The screenshots for the other treatment order are identical, except that Parts One and Two are swapped.

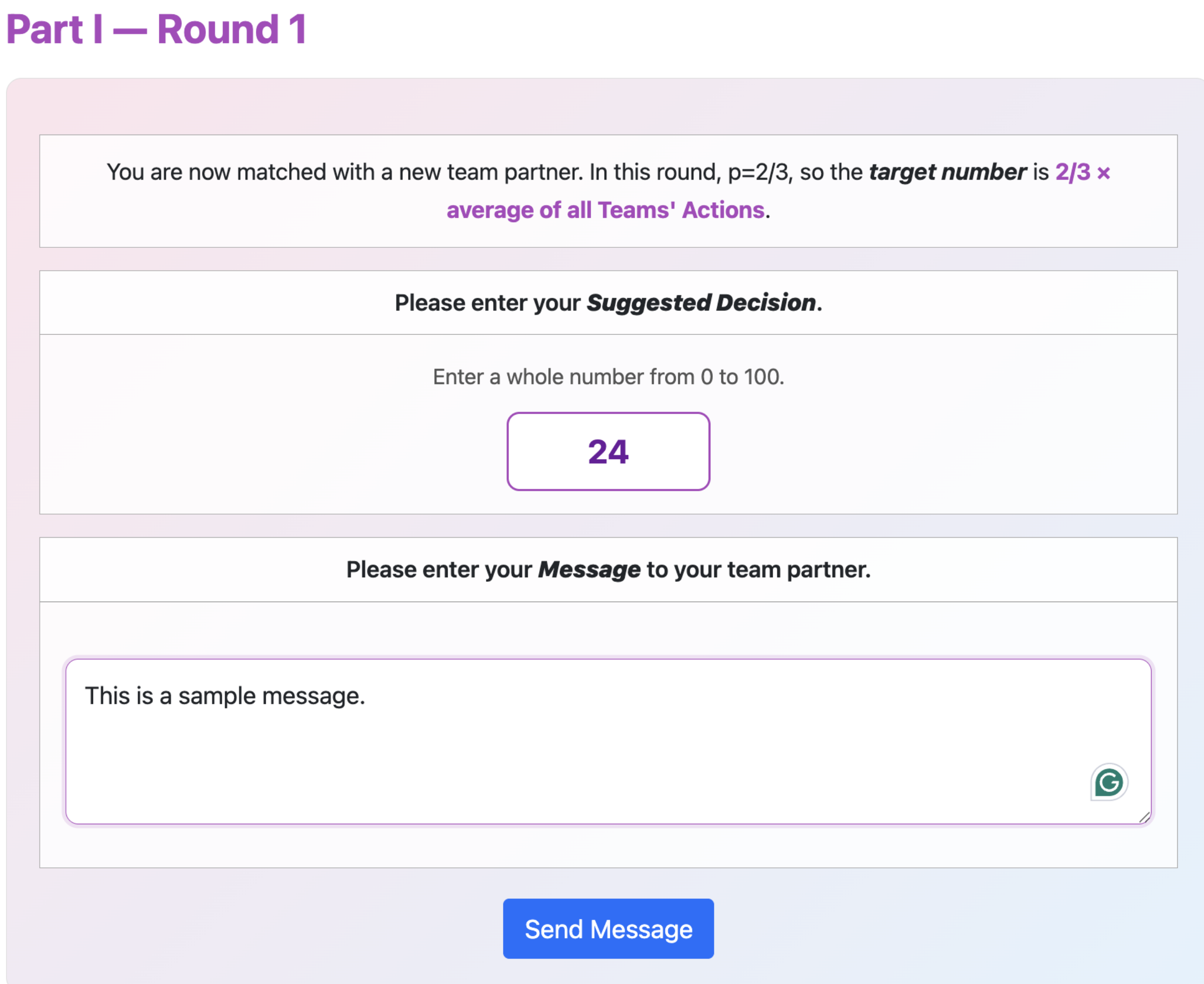

Figure A2: Messaging Stage of the Beauty Contest Game

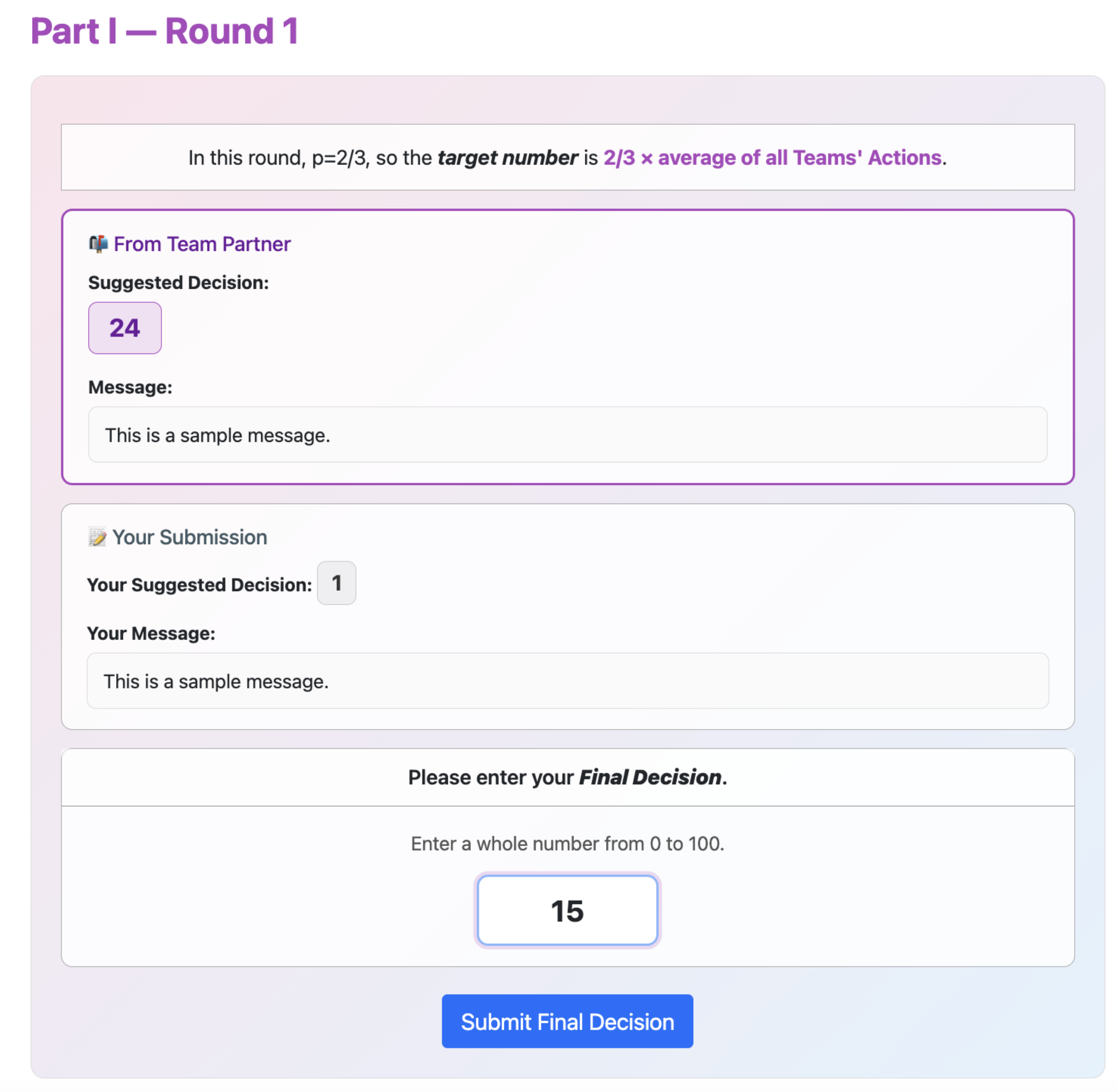

Figure A3: Decision Stage of the Beauty Contest Game

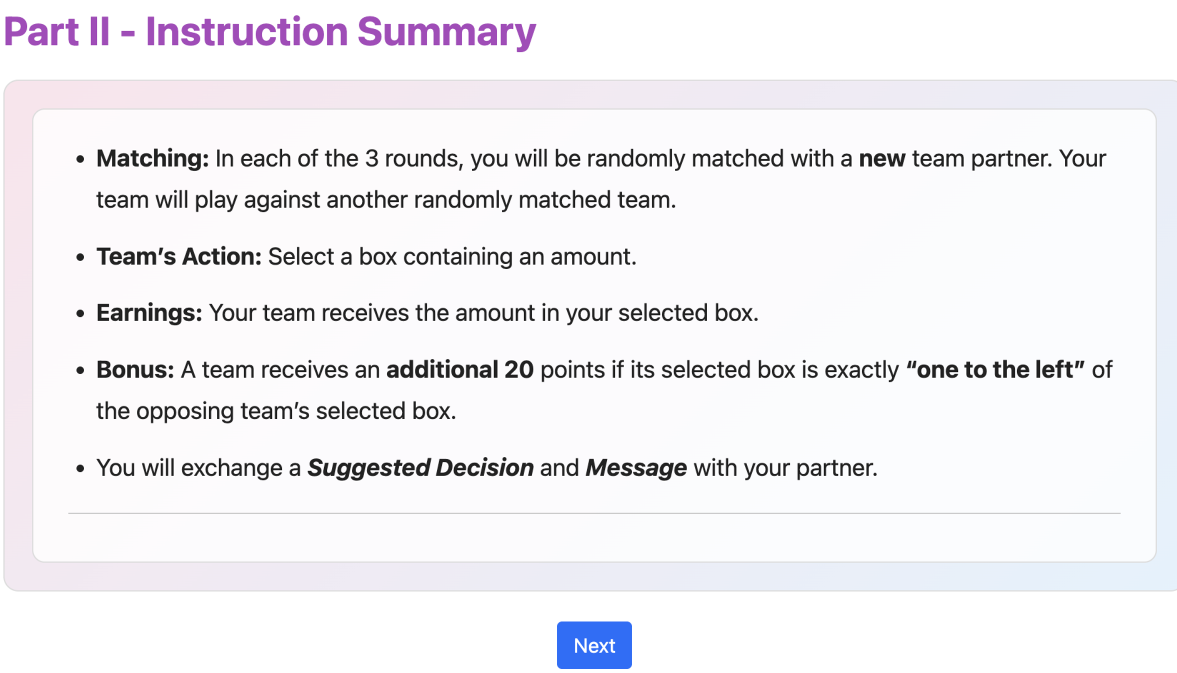

Figure A4: Instruction Summary of the 11-20 Game

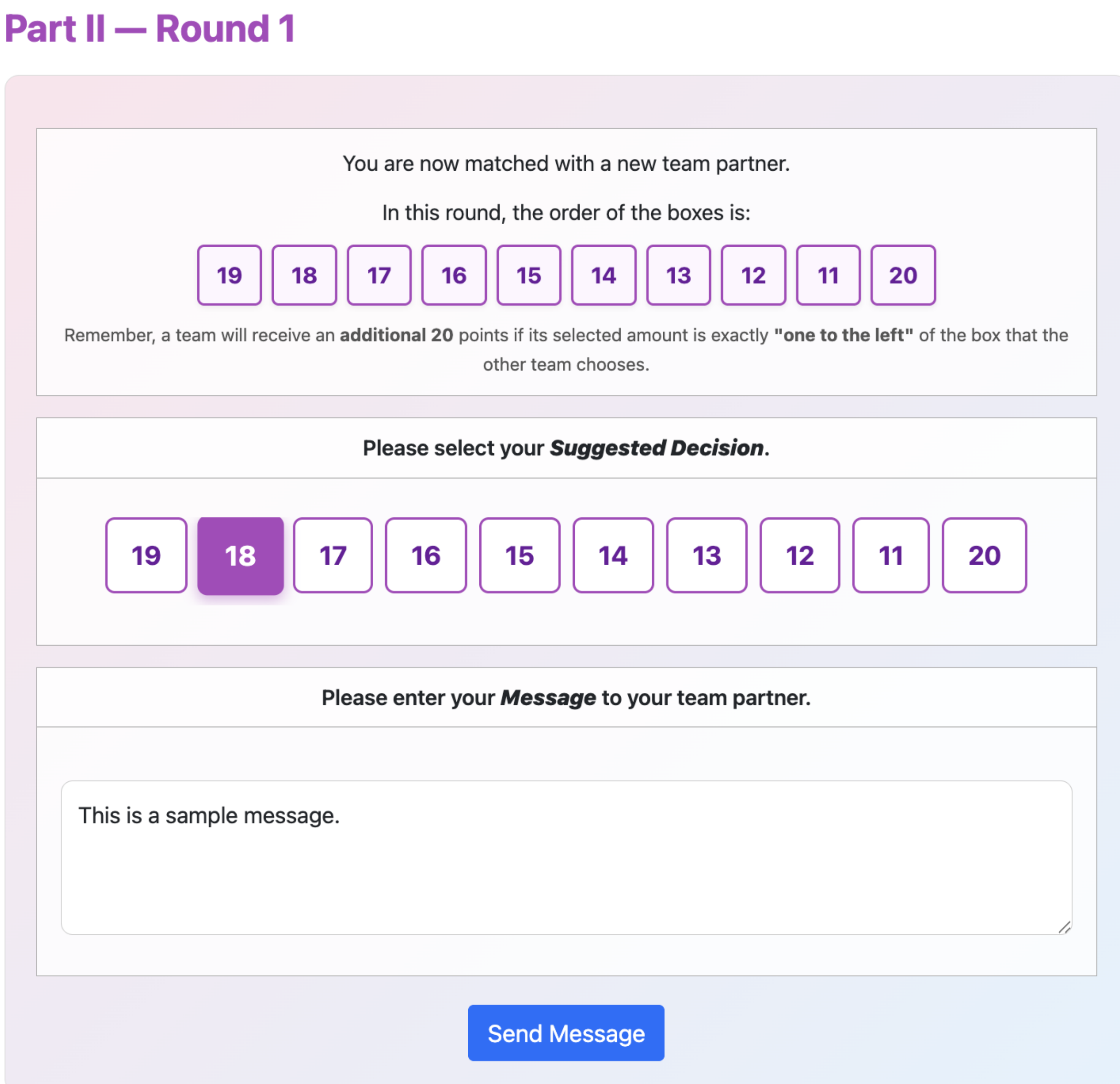

Figure A5: Messaging Stage of the 11-20 Game

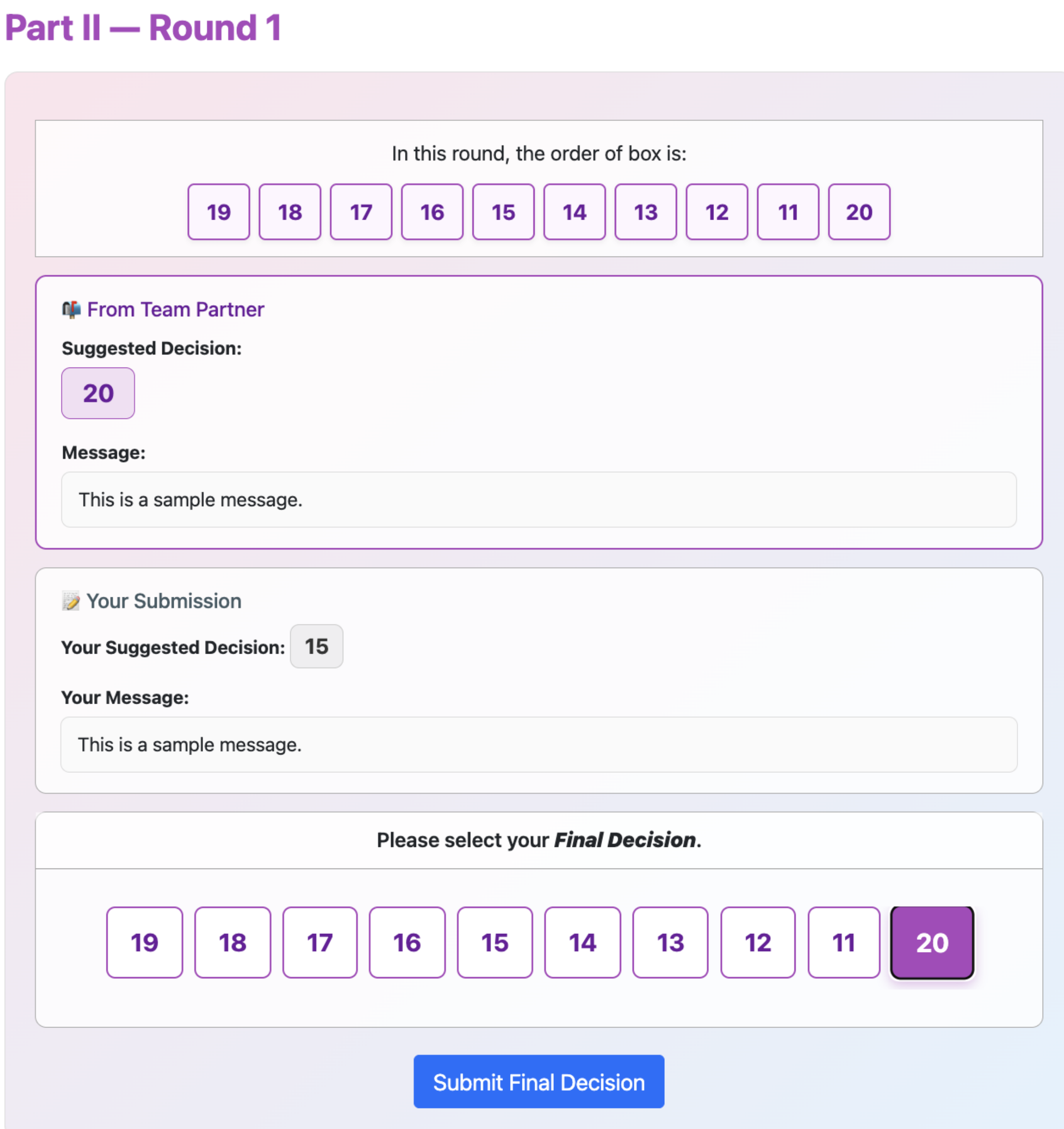

Figure A6: Decision Stage of the 11-20 Game

Your Earnings Summary

Experiment Complete — Earnings Summary

Below is a detailed record of all six rounds. Your total earnings are shown at the bottom.

Part I Results

Round	Team's Action	Target Number	Winning Team's Action	Team Earnings	Individual Earnings
Round 1 (p = 2/3)	**7**	6.33	7	**100**	**50.0**
Round 2 (p = 1/3)	**0**	2.0	0	**100**	**50.0**
Round 3 (p = 1/2)	**30**	11.25	15	**0**	**0.0**
				Part I Subtotal	**100.0**

Part II Results

Round	Team's Action	Opposing Team's Action	Bonus?	Team Earnings	Individual Earnings
Round 1	**11**	20	Yes (20)	**31.0**	**15.5**
Round 2	**13**	16	No	**13.0**	**6.5**
Round 3	**20**	19	No	**20.0**	**10.0**
				Part II Subtotal	**32.0**

Total points earned (all 6 rounds)	132.0 pts
Game earnings (132.0 pts ÷ 4)	$33.00
Show-up fee	$10.00

Your total earnings today

$43.00

Please remain seated. The experimenter will collect your earnings shortly.

Figure A7: Feedback and Earnings Summary